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CHE 312 Transport Phenomena II
Instructor Vivek Sharma

Name EWA GOSEWSKI

Quiz 1: Living and Working in Chiberia

(Original problems: do not repost anywhere without permission)
(Recall the honor code & trust your own instincts & knowledge)
Marked out of 100; contribution to final grade varies.

After four years of chemical engineering education, you find yourself being interviewed by a major chemical engineering consulting firm for a dream job with a dream salary at a dream location. You have no idea what to expect in your interview, and you go in, thinking, I know what I know, and I will do as well as I can.

A series of questions follow. The hiring engineer has to take 100 interviews in a week, and would like to get quick, crisp answers.

(a) What are the definitions and S. I. units for **q** (heat flux), k (thermal conductivity), $\hat{C}_p$ (heat capacity per unit mass), thermal diffusivity, stress and shear viscosity? (Correct formula will earn you complete credit for the definition, and correct units for units). (24)

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heat flux , [W/m²] $q_y = \frac{-k\,dT}{dy}$

thermal conductivity k

$\hat{C}_p$ heat capacity [J/kg K] $\hat{C}_p = \frac{k}{\rho\alpha}$ $C_p = \frac{Pr\, k}{\mu}$

thermal diffusivity $\alpha = \frac{k}{\rho \hat{C}_p}$ [m²/s]

stress $\tau_{yx} = -\mu \frac{dv_x}{dy}$ [Pa]

shear viscosity [Pa·s] $\mu = \tau \cdot t$ [Pa]

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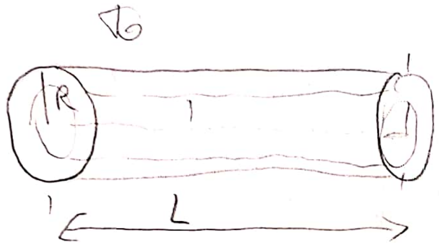
(b) A client's company has postulated that their design of electrical wire will help in saving transmission losses, and you are just trying to calibrate the wire using standard heat transfer calculations. Consider an electrically heated cylindrical wire of radius R, that loses heat convectively to the outside air at temperature T_0 with a heat transfer coefficient h, has a contant heat production rate per unit volume S_e. You can pick a length L for the sake of your computation. Determine $T(r)$ assuming temperature rise is small enough that both thermal conductivity and electrical conductivity have a constant value. Find the heat outflow from the surface of the wire. (Show your work. Draw a schematic. Specify assumptions, and boundary conditions). (40)



GIVEN: R = r
To
h
constant heat production rate / Se
T(r) = ?

$IN - OUT + PROD = \cancel{ACC}^{0}$ $\qquad Se = \dfrac{I^2}{ke}$

$IN = \underbrace{q_R|_r}_{FLUX} \cdot \underbrace{2\pi r L}_{AREA}$

$OUT = q_R|_{R+\Delta R} \cdot 2\pi(R+\Delta r)L$

$PROD = Se \cdot \underbrace{2\pi R \Delta R L}_{Volume}$

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$\lim_{\Delta R \to 0} \dfrac{R q_R|_{R+\Delta R} - R q_R|_r}{\Delta r} \Rightarrow R q_R = \dfrac{Se R^2}{2} + C$

$\dfrac{d}{dr} r q_r = r Se \Rightarrow \int q_r \, dr = \int r \, Se \, dr.$

$dr \; r q_r = \dfrac{r^2}{2} Se + C_1$

B.C.1 $r = 0$ $\quad C_1 = 0$ $\quad q_r = \dfrac{r Se}{2}$

$-\dfrac{k dT}{dr} = \dfrac{r Se}{2} \Rightarrow \int dT = \int \dfrac{r Se \, dr}{-2k}$

$T(r) = \dfrac{-r^2}{4k} + C_2$

B.C.2 $r = R$, $T = T_0$ $= -\dfrac{R^2 Se}{4k} + C_2$

$\Rightarrow C_2 = \dfrac{R^2 Se}{4k} + T_0 \Rightarrow$

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(c) Find the maximum temperature in a wire if the temperature profile is known to be $T(r) = T_o\left(2r/3R - (r/3R)^3\right)$ at a steady state, assuming T_o of 20 degree Celsius, and the diameter of the ~~fish~~ wire is 4 cm. (16) $R = 2$

$f'(r) = 0$

$f''(r) < 0$

$$T(r) = \frac{20 \cdot 2r}{3(2)} - \left(\frac{20r}{6}\right)^3$$

$$T(r) = \frac{\cancel{20}\ 40}{3} r - \frac{10}{3} r^3$$

$$T'(r) = \frac{20}{3} + 10r^2$$

$$\frac{20}{3} + 10r^2 = 0 \Rightarrow$$

$$r^2 = \frac{1}{3}$$

$$r = \sqrt{\frac{1}{3}}$$

$$f''(r) = 20r$$

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(d) Consider steady state, laminar flow of a fluid of constant density, ρ and viscosity, μ in a vertical tube of length L and radius R. (a) Show the velocity profile within the tube. (b) Write the Hagen-Poiseuille equation. (c) To pump the fluid at the same flow rate, but through a channel twice in radius, requires a pressure drop that is X times higher or lower. Find X and specify if it is higher or lower. (7+7+6)

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